Math 211 **Summer 2022**
Quiz 1 **Drew Armstrong**

Problem 1. Parametrized Paths. Consider the following parametrized path in $\mathbb{R}^2$:

$$f(t) = (x(t), y(t)) = (1 + 3t^2, 4t^2).$$

(a) Compute the velocity vector $f'(t)$ and the speed $\|f'(t)\|$.
(b) Compute the arc length between $t = 0$ and $t = 1$.

Problem 2. Vector Arithmetic. Consider the triangle in $\mathbb{R}^3$ with vertices

$$P = (1, 1, 1), \quad Q = (2, -1, 1), \quad R = (1, 2, 3).$$

(a) Compute the cosines of the angles of the triangle. [No need to find the actual angles.]
(b) Find the equation of the plane in $\mathbb{R}^3$ that contains this triangle.